

1 Test-procedure choice for the biomarker-treatment interaction
2 in aggregated N-of-1 trials: dichotomization, classical
3 RM-ANOVA, mixed-effects, and generalized estimating
4 equations

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32 1 Abstract

33 **Background.** Inference about the biomarker-treatment interaction in aggregated N-of-1
34 trials depends on the analyst’s test procedure: classical repeated-measures ANOVA (RM-
35 ANOVA), the linear mixed-effects model (LME), or generalized estimating equations (GEE). A
36 common but rarely examined feature of the classical analysis is that it requires the continuous
37 biomarker to be dichotomized. We quantify the consequences of test-procedure choice for the
38 interaction, separating the cost of dichotomization from the cost of the test procedure proper.

39 **Methods.** We derive the closed-form strict RM-ANOVA F -statistic for the biomarker-
40 stratified treatment-by-time interaction in a balanced split-plot design with categorical pre-
41 dictors and the sphericity assumption, and relate it to the Wald t -test for the corresponding
42 fixed-effects coefficient in a linear mixed model with continuous-time AR(1) residual correla-
43 tion. We then conduct a Monte Carlo simulation study, designed and reported under the¹
44 ADEMP framework, that crosses five test procedures (strict RM-ANOVA on the dichotomized
45 biomarker; a continuous-biomarker within-subject contrast; the standard pmsimstats linear-
46 mixed analysis with `corCAR1`; and GEE with naive and with Mancl-DeRouen small-sample
47 sandwich variance) against the biomarker-moderation coefficient $\beta_{bm} \in \{0, 0.45\}$ and sample
48 size $N \in \{25, 35, 70, 100\}$, a 40-cell factorial at the prazosin-PTSD reference design with 1000
49 replicates per power cell and 2000 per Type I cell. The continuous-biomarker contrast dif-
50 fers from strict RM-ANOVA only in retaining the continuous biomarker, isolating the cost of
51 dichotomization. The cycle-by-period design grid is specified here but its empirical sweep is
52 deferred to a companion paper.

53 **Results.** The closed-form strict RM-ANOVA test agrees with the linear-mixed Wald test
54 under sphericity and balanced design. In the simulation, most of the power gap between
55 strict RM-ANOVA and the mixed model is the cost of dichotomization rather than of the test
56 procedure: at $N = 25$, replacing the dichotomized biomarker with the continuous one raises
57 power by 0.180, while the further step to the full mixed model adds only 0.027, a pattern
58 stable across N (dichotomization 0.08–0.21, test procedure at most 0.04). Among the four
59 continuous-biomarker procedures, the linear-mixed analysis attains the highest power within
60 nominal Type I error; GEE with the naive sandwich inflates Type I error to between $1.5\times$ and
61 $1.9\times$ nominal across all four sample sizes, which the Mancl-DeRouen correction removes at
62 a moderate (about 10-percentage-point) power cost while restoring 93–95% interval coverage.
63 The inline Mancl-DeRouen standard error agrees with the `geesmv` reference to within 2.5%.

64 **Conclusions.** The analyst’s most consequential decision is to keep the biomarker contin-
65 uous rather than dichotomize it; among continuous-biomarker procedures the linear-mixed
66 analysis is the preferred default and a dichotomization-free within-subject contrast is a close,
67 transparent alternative. Naive GEE should not be used at the sample sizes examined without

68 a small-sample correction. This paper resolves the test-procedure axis at a fixed reference
69 design; the joint cycle-by-period design grid is specified here and deferred to a companion
70 paper.

71 2 Introduction

72 2.1 A claim about how design and test choices interact

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75 The present paper develops the methodological case for treating the analyst's two
76 principal choices in an aggregated N-of-1 trial jointly rather than separately. The
77 two choices are the test procedure for the biomarker-treatment interaction (strict
78 repeated-measures ANOVA, the linear mixed-effects model with continuous-time
79 AR(1) residual correlation, the mixed-effects model for repeated measures, gener-
80 alized estimating equations) and the trial-design parameters that govern within-
81 subject treatment cycling (number of treatment cycles, on-drug and off-drug pe-
82 riod lengths, on/off symmetry, optional run-in and run-out phases). The validity
83 and power of any given test depend on what the design assumes about carryover,
84 autocorrelation, and within-subject variance structure; conversely, the optimal de-
85 sign parameters depend on which test the analyst plans to use. Treating the two
86 as independent optimizations, as the published methodological literature typically
87 does, produces locally optimal but jointly suboptimal combinations, and we shall
88 argue that this gap is consequential at the sample sizes characteristic of precision-
89 medicine pilot studies.

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92 The case we develop is two-part. We begin by establishing that a joint design-by-
93 test sensitivity study has not, to the best of our knowledge, been performed for the
94 aggregated N-of-1 design class, and that the absence of one limits the field's ability
95 to make defensible joint recommendations. We then demonstrate, through a fo-
96 cused Monte Carlo experiment, that the linear mixed-effects test with continuous-
97 time AR(1) residual correlation, used in conjunction with the Hendrickson hybrid
98 open-label-plus-blinded-discontinuation design², dominates the strict classical RM-
99 ANOVA test in essentially every parameter cell where within-participant residuals
100 depart from compound symmetry. That is, strict RM-ANOVA remains useful as
101 a stress test — the most stringent baseline that compound-symmetry machinery
102 can deliver — but not as a recommendation in its own right.

103 2.2 Why the question matters for chronic-condition pharmacology

104 In this section we shall set out three reasons why the joint treatment of test procedure and
105 design parameters is warranted specifically for the aggregated N-of-1 design class.

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108 We begin with the physiological motivation. Within-participant residuals in
109 chronic-condition trials are autocorrelated not as an analytic convenience but
110 because most chronic-condition response trajectories exhibit serial correlation on
111 a timescale set by drug pharmacokinetics, condition-specific natural-history drift,
112 and measurement reactivity (Hawthorne, structured-visit, and diary-completion
113 effects). Continuous-time AR(1) residual correlation is, in our view, the appro-
114 priate structural prior for this physiology, but it does not satisfy the sphericity
115 assumption required by strict RM-ANOVA.³ show through simulation that un-
116 corrected RM-ANOVA inflates Type I error under sphericity violations; they also
117 report, importantly, that the Greenhouse-Geisser-corrected RM-ANOVA and the
118 multilevel linear model perform comparably once the correction is applied, and
119 that the multilevel model is not uniformly safer (it can be the more liberal of the
120 two in some conditions). The degrees-of-freedom corrections^{4,5} recover nominal
121 levels only conditionally.⁶ extend this evidence with finer epsilon resolution, with
122 the conventional rule favoring Greenhouse-Geisser when $\epsilon < 0.75$. The advantage
123 we claim for the mixed-effects model with explicit `corCAR1` residual structure is
124 thus not that it is uniformly more powerful than a well-corrected RM-ANOVA,
125 but that it models the AR(1) structure directly, accommodates the continuous
126 biomarker and unequal time spacing, and so avoids both the dichotomization and
127 the degrees-of-freedom penalty that the classical route incurs.

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130 The biomarker-treatment interaction also merits separate attention because
131 it is harder to detect than the average treatment effect (ATE). The PATH
132 framework^{7,8} makes the methodological distinction precise: effect-modeling
133 (the predictive heterogeneity-of-treatment-effects, or HTE, question) requires
134 more design-aware machinery than risk-modeling (the ATE question). The
135 within-participant contrasts that contribute to the interaction estimator are
136 sensitive to the trial-design parameters in ways the ATE analysis is not, and a
137 test procedure that underpowers the interaction term will systematically miss
138 predictive-biomarker signal even at sample sizes that appear adequate for the
139 ATE test.

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142 Finally, cycle count and period length jointly determine the within-subject con-
143 trast structure in ways that cannot be optimized independently.⁹ and¹⁰ establish,
144 in the broader cross-over literature, that adding within-subject treatment cycles
145 contributes information about the patient-by-treatment interaction variance that
146 no cross-sectional comparator can supply. The N-of-1 specialization of this princi-
147 ple is that more cycles increase power for the biomarker-treatment interaction, but

148 with diminishing returns: cycles beyond the fourth or fifth contribute marginal
149 information at the cost of trial duration. Period length interacts with carryover
150 — shorter periods produce more cycles but worse separation between on-drug and
151 off-drug states once carryover is non-trivial. The joint optimum on the (cycles, pe-
152 riod length) plane depends on the carryover half-life and the chosen test procedure
153 together, not on each factor in isolation.

154 **2.3 What the published N-of-1 methodology literature does and does not** 155 **do**

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158 We begin with the state of the analytic-model literature. The systematic re-
159 view of¹¹ documents that 83.8% of the published N-of-1 studies they catalogue
160 ignore autocorrelation entirely;¹² and¹³ reach similar conclusions in neurology and
161 behavioral-medicine subdomains. CONSORT extension reporting standards^{14,15}
162 codify reporting quality but do not address the analytic-model decision in any de-
163 tail, and¹⁶ identifies the analytic-method gap as the highest-priority methodologi-
164 cal problem in the field. It appears, then, that the majority of N-of-1 analyses are
165 conducted under a misspecified residual covariance assumption, and that the field
166 has not yet converged on remedies.

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169 Test-procedure comparisons for longitudinal data are not new. The generic con-
170 trast of RM-ANOVA against GEE and mixed-effects models is well established:¹⁷,
171 for instance, report through simulation that RM-ANOVA is appreciably less power-
172 ful than GEE and mixed models for continuous longitudinal endpoints, the same
173 ordering we recover. What that literature does not do is treat the biomarker-
174 treatment interaction estimand, the aggregated N-of-1 / cross-over design, or the
175 continuous-time AR(1) structure together, nor does it isolate the dichotomization
176 cost that the classical route imposes specifically when the moderator is continuous.
177 The MMRM consensus established by Mallinckrodt and colleagues^{18–21} applies to
178 parallel-group longitudinal trials with monotone missingness, not to aggregated
179 N-of-1 designs. The GEE methodology of²² with sandwich variance estimation
180 has been adapted to small-cluster settings via bias-corrected estimators^{23,24}, but
181 the small-sample concerns are acute at trial-relevant N-of-1 cluster counts:²⁵ re-
182 port that in stepped-wedge cluster trials with fewer than six independent units
183 only 6.6% of analyses use appropriate small-sample corrections, illustrating the
184 gap between method availability and practice. None of these papers address the
185 biomarker-interaction estimand in the cross-over or N-of-1 setting directly.

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Design-parameter optimization has been studied, but separately from test-procedure choice.²⁶ develop power calculations for $(AB)^k$ designs that match the N-of-1 within-subject contrast structure;²⁷ document strong autocorrelation effects on permutation-test power; and¹⁰ demonstrates how replicate cross-over designs allow direct estimation of patient-by-treatment interaction variance, the conceptual ancestor of N-of-1 aggregation. The Hendrickson et al. framework² is, to the best of our knowledge, the only paper that compares full design families head-to-head for the biomarker-interaction power question, but it holds period length fixed and uses a single test procedure.²⁸ compare aggregated N-of-1 trials against parallel and cross-over designs by simulation and find the aggregated design more efficient, but analyze a single mixed model and do not vary the test procedure.²⁹ and³⁰ synthesize the broader N-of-1 design literature without addressing the joint design-by-test sensitivity question.

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The gap, then, is this. Across the recent systematic reviews^{11,12,31} and the methodological literature surveyed above, no published study crosses test procedure (strict RM-ANOVA, MMRM, GEE, mixed-effects with `corCAR1`) against a design-parameter grid (cycles, period lengths, asymmetry, run-in, run-out) under simulation for the aggregated N-of-1 design class. The present paper endeavors to address that gap, at least on the test-procedure axis.

2.4 Run-in design and the placebo-selection controversy

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A separate methodological controversy that the joint design-by-test framing intersects concerns the use of placebo run-in phases to exclude placebo responders before randomization.³² show formally that response-based exclusion during run-in introduces a selection bias that damages external validity and may affect internal validity as well. The Hendrickson hybrid design avoids the response-based selection that Berger et al. flag by using an open-label stabilization phase rather than a placebo run-in; the design choice is therefore conservative on the run-in question, though it forgoes the power that aggressive response-based selection would otherwise supply. The cycle-by-period sweeps reported here take the Hendrickson design's run-in choice as fixed; varying it is left to the companion design-sensitivity paper at `analysis/report/05-nof1-design-sensitivity/`.

2.5 What this paper contributes

We make four contributions, each addressed in a section below.

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First, we derive the closed-form strict RM-ANOVA F -statistic for the biomarker-stratified treatment-by-time interaction in a balanced split-plot design with categorical predictors and the sphericity assumption (§2). We relate the strict-RM-ANOVA test to the Wald t -test on the biomarker-treatment fixed-effects coefficient in a linear mixed model with continuous-time AR(1) residual correlation, identifying the conditions on residual covariance under which the two tests agree and the regimes in which they diverge. The derivation extends standard cross-over methodology^{9,33} to the biomarker-stratified case and clarifies the connection to the contemporary sphericity-violation literature^{3,6}.

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Second, we specify a cycle-by-period design grid (§3) that varies the number of treatment cycles, on-drug and off-drug period lengths, and on/off symmetry around the Hendrickson hybrid reference. The grid is calibrated to the prazosin/PTSD reference application that motivates the pmsimstats program.

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Third, we conduct a Monte Carlo simulation study (§4–§5) that crosses five test procedures (strict RM-ANOVA on the dichotomized biomarker; a continuous-biomarker within-subject contrast; the standard pmsimstats linear-mixed analysis with corCAR1; and GEE with naive and with Mancl-DeRouen³⁴ small-sample sandwich variance) against the biomarker-moderation coefficient and sample size at the fixed reference design. The continuous-biomarker contrast is the dichotomization-free analogue of the strict RM-ANOVA test and lets us separate the cost of dichotomizing the biomarker from the cost of the test procedure proper. The simulation runs on the pmsimstats package data-generating process under the mean-moderation architecture via `generateData()` and `lme_analysis()`, holding fixed the response-curve and covariance-structure axes addressed in `analysis/report/07-gompertz-evaluation/` and `analysis/report/01-dgp-mean-moderation-vs-mvn/`. Results are reported in the¹ ADEMP framework adopted as the project-wide simulation-reporting standard.

Notation for the moderation parameter. Because the data-generating process here is the mean-moderation architecture throughout, the biomarker-moderation parameter is written β_{bm} , the dimensionless multiplier scaling the treatment effect by the standardized biomarker. The companion architecture manuscript reserves c_{bm} for the Architecture-B parameter, which is a correlation rather than a regression multiplier. The two are calibrated to a common numeric scale (the Architecture-A shift is $\beta_{bm} b_i \sigma_{BR}$ on the standardized biomarker b_i), so the reference value 0.45 denotes a matched moderation strength under either architecture, but the symbols are not interchangeable.

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270 Fourth, we synthesize a design-and-test recommendation table (§6) for typical
271 chronic-condition N-of-1 trial scenarios. The recommendation is presented as a ta-
272 ble of preferred (cycle count, period length, test procedure) combinations indexed
273 by the carryover half-life and the target effect size, rather than as a single ‘best’
274 design.

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277 The remainder of the paper is organized as follows. Section 2 develops the strict
278 RM-ANOVA derivation, adapted from docs/26-rm-anova-interaction-test.md.
279 Section 3 specifies the cycle-by-period design grid, adapted from docs/27-cycle-period-design-sweep-p
280 Section 4 lays out the simulation design. Section 5 reports the results. Section
281 6 synthesizes the joint design-and-test recommendation. Section 7 discusses the
282 implications for the methodology of N-of-1 biomarker-validation trials and for the
283 operating characteristics of the pmsimstats simulation program.

284 3 Closed-form strict RM-ANOVA test of the biomarker- 285 treatment interaction

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288 In this section we shall derive the closed-form strict RM-ANOVA F -statistic for
289 the biomarker-treatment interaction in a balanced split-plot design with categori-
290 cal predictors and the sphericity assumption, and relate it to the Wald t -statistic
291 on the corresponding linear-mixed coefficient. The derivation specializes classical
292 split-plot machinery³⁵ to the biomarker-stratified case in the simplest design that
293 supports the test: two biomarker strata, two treatment phases, and T within-
294 phase timepoints. The presentation is deliberately confined to strict RM-ANOVA.
295 Modern extensions — the mixed-effects model for repeated measures³⁶, general-
296 ized estimating equations²², and ANCOVA with random subject effects — appear
297 only for context; the closed-form expression we report is the textbook compound-
298 symmetric form. Three findings anchor the rest of the paper:

- 299 1. Under sphericity and balance, the strict RM-ANOVA test of the biomarker-treatment
300 interaction is identical to a two-sample t -test on the within-subject treatment differences,
301 with degrees of freedom $2(n - 1)$.
- 302 2. The closed form yields a clean non-central F power formula that we use as the analytic
303 anchor for the simulation study in §4-§5.
- 304 3. Five assumptions of the closed form are violated by the pmsimstats data-generating
305 process (DGP; continuous biomarker, AR(1) within-subject covariance, exponential car-
306 rryover, missing data, unequal time spacing). Each violation favors the linear-mixed

analysis over the strict RM-ANOVA test in operating characteristics, motivating the primary-and-sensitivity reporting strategy of §6.

3.1 Notation and design

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We begin with the simplest design that supports a biomarker-treatment interaction test in a within-subject setting. The reader will note that the design structure described here is more restrictive than the pmsimstats DGP; we return to that gap explicitly in §2.4.

Specifically, we consider the following structural elements:

- Two biomarker strata $i \in \{1, 2\}$, formed by dichotomizing the continuous biomarker at a pre-specified cut-point. Strict RM-ANOVA requires categorical predictors; the dichotomization is the price of admission to the framework.
- n subjects per stratum, balanced. Index $l \in \{1, \dots, n\}$.
- Two treatment phases $j \in \{1, 2\}$, active drug and placebo. In an N-of-1 design these are two phases through which every subject passes; in a parallel-group design they are between-subject and the design is not a split-plot.
- T within-phase timepoints, indexed $k \in \{1, \dots, T\}$. In the simplest case $T = 1$, the analyst pre-averages within-phase observations to a single mean per subject per phase.

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The biomarker stratum is between-subjects (each subject sits in exactly one stratum); the treatment phase and the timepoints are within-subjects (each subject sees both phases at every timepoint). This is the canonical split-plot repeated-measures design³⁵.

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We write Y_{ijkl} for the symptom outcome at biomarker stratum i , treatment phase j , subject l within stratum i , and timepoint k . The strict classical split-plot model is

$$Y_{ijkl} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \pi_{l(i)} + \tau_k + (\alpha\tau)_{ik} + (\beta\tau)_{jk} + (\alpha\beta\tau)_{ijk} + e_{ijkl}, \quad (1)$$

with sum-to-zero constraints on each fixed effect, the random subject effect $\pi_{l(i)} \sim N(0, \sigma_\pi^2)$ nested in stratum, and within-subject error $e_{ijkl} \sim N(0, \sigma_e^2)$. The model presupposes the classical sphericity (compound symmetry) condition on the within-

subject covariance: every pair of within-subject observations has the same covariance σ_π^2 , and every observation has the same total variance $\sigma_\pi^2 + \sigma_e^2$.

3.2 Sums of squares and the interaction F -statistic

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We begin the derivation by partitioning the total sum of squares. The total sum of squares partitions into between-subjects and within-subjects components:

$$SS_{\text{tot}} = SS_{\text{between}} + SS_{\text{within}}. \quad (2)$$

The between-subjects component decomposes further into SS_A (biomarker stratum) and $SS_{S(A)}$ (subject within stratum):

$$SS_A = bnT \sum_{i=1}^a (\bar{Y}_{i...} - \bar{Y}_{...})^2, \quad SS_{S(A)} = bT \sum_{i=1}^a \sum_{l=1}^n (\bar{Y}_{i.l} - \bar{Y}_{i...})^2. \quad (3)$$

The within-subjects component decomposes into SS_B (treatment phase), SS_{AB} (the biomarker-treatment interaction, the primary target of the test), and a within-subjects error term $SS_{B \times S(A)}$:

$$SS_{AB} = nT \sum_{i,j} (\bar{Y}_{ij..} - \bar{Y}_{i...} - \bar{Y}_{.j..} + \bar{Y}_{...})^2, \quad (4)$$

$$SS_{B \times S(A)} = T \sum_{i,j,l} (\bar{Y}_{ijl.} - \bar{Y}_{i.l} - \bar{Y}_{ij..} + \bar{Y}_{i...})^2. \quad (5)$$

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Corresponding degrees of freedom are $df_A = a - 1$, $df_{S(A)} = a(n - 1)$, $df_B = b - 1$, $df_{AB} = (a - 1)(b - 1)$, and $df_{B \times S(A)} = a(n - 1)(b - 1)$, the last being the within-subjects error degrees of freedom appropriate for the treatment effect and its interaction with the between-subjects factor. The biomarker-treatment interaction is then tested by the F -ratio

$$F_{AB} = \frac{MS_{AB}}{MS_{B \times S(A)}} = \frac{SS_{AB}/df_{AB}}{SS_{B \times S(A)}/df_{B \times S(A)}}. \quad (6)$$

Under the null $H_0 : (\alpha\beta)_{ij} \equiv 0$ and the sphericity assumption, $F_{AB} \sim F(df_{AB}, df_{B \times S(A)})$.

3.2.1 The 2×2 split-plot case

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369 We now specialize to $a = b = 2$ (two biomarker strata, two phases), setting $T = 1$
 370 for cleanest exposition; that is, the analyst pre-averages within-phase observations
 371 to a single value per subject per phase before computing the interaction. The
 372 sample interaction contrast is

$$373 \quad \hat{L} = (\bar{Y}_{11\cdot} - \bar{Y}_{12\cdot}) - (\bar{Y}_{21\cdot} - \bar{Y}_{22\cdot}), \quad (7)$$

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376 That is, $\hat{L}$ is the difference of within-subject treatment differences across biomarker
 377 strata, and is the natural biomarker-treatment interaction statistic. The interac-
 378 tion parameter under the sum-to-zero ANOVA parameterization satisfies $(\widehat{\alpha\beta})_{11} =$
 379 $\hat{L}/4$ (and the other three cells take values $\pm\hat{L}/4$), so the interaction sum of squares
 380 is $SS_{AB} = nT\hat{L}^2/4$, and with $df_{AB} = 1$ this equals the interaction mean square.
 381 Under sphericity and balance, $MS_{B \times S(A)} = \hat{\sigma}_e^2$. Substituting into equation (6)
 382 yields

$$383 \quad F_{AB} = \frac{nT\hat{L}^2}{4\hat{\sigma}_e^2}, \quad df_1 = 1, \quad df_2 = 2(n-1). \quad (8)$$

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386 Equation (8) is the closed-form expression for the strict classical RM-ANOVA test
 387 statistic for the biomarker-treatment interaction in a $2 \times 2 \times T$ split-plot design.
 388 We shall refer to it as the ‘strict RM-ANOVA F ’ throughout the rest of the paper,
 389 to distinguish it from the mixed-effects and GEE Wald statistics derived from the
 390 same data.

391 **3.2.2 Equivalence to a within-subject t -test**

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394 We note that equation (8) is equivalent to a two-sample t -test on the within-subject
 395 differences. Define $\Delta_{il} = \bar{Y}_{i1l} - \bar{Y}_{i2l}$ as the within-subject treatment difference for
 396 subject l in biomarker stratum i , averaged over T within-phase timepoints. Then

$$397 \quad \hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2, \quad \bar{\Delta}_i = \frac{1}{n} \sum_{l=1}^n \Delta_{il}. \quad (9)$$

398 Under sphericity, Δ_{il} are i.i.d. within stratum with $\text{Var}(\Delta_{il}) = 2\sigma_e^2/T$, since the random
 399 subject effect cancels in the within-subject difference. The variance of $\hat{L}$ is therefore

$$\text{Var}(\hat{L}) = \frac{2 \cdot 2\sigma_e^2}{nT} = \frac{4\sigma_e^2}{nT}, \quad (10)$$

and the t -statistic for the contrast is

$$t_{AB} = \frac{\hat{L}}{\sqrt{4\hat{\sigma}_e^2/(nT)}} = \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_e}, \quad (11)$$

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We see that $t_{AB}^2 = F_{AB}$, recovering equation (8). The strict-RM-ANOVA F -test for the biomarker-treatment interaction is therefore identical to a two-sample t -test on the within-subject treatment differences, with degrees of freedom $2(n-1)$. This identity is well-known in the split-plot literature³⁵ and is the analytical anchor for the statement that the strict-RM-ANOVA test of a biomarker-treatment interaction reduces to a simple two-sample comparison of treatment-effect estimates between biomarker strata.

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That is, under sphericity and balance, the biomarker-treatment interaction F -test asks simply whether the within-subject treatment effect differs between biomarker strata, and the answer is given by a two-sample t -test on those within-subject differences. This reduction has intuitive appeal and makes the test's assumptions unusually transparent: anything that biases or inflates the within-subject differences will distort the test in a predictable direction. We shall exploit this transparency when relating the strict-RM-ANOVA results to the mixed-effects results in §5.

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For a biomarker strata, b treatment phases, n subjects per stratum, and T within-phase timepoints, equation (6) generalizes directly with degrees of freedom $df_1 = (a-1)(b-1)$ and $df_2 = a(n-1)(b-1)$. Including the time factor τ_k explicitly, the test of whether the biomarker modifies the treatment effect across time is the three-way interaction $(\alpha\beta\tau)_{ijk}$; the corresponding F -statistic is $F_{AB\tau} = MS_{AB\tau}/MS_{B\tau \times S(A)}$ with $df_1 = (a-1)(b-1)(T-1)$ and $df_2 = a(n-1)(b-1)(T-1)$.

3.3 Relation to the linear-mixed Wald test

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We shall now relate the strict-RM-ANOVA F -statistic of equation (8) to the `pmsimstats nlme::lme` Wald t -statistic on the `bm:Dbc` coefficient. The two are asymp-

totically equivalent under three conditions: (i) the biomarker is dichotomized at the cut-point that defines the strata, (ii) the within-subject covariance is exactly compound-symmetric, and (iii) Dbc is binary, the E1 specification of the carryover-sensitivity manuscript rather than the matched-decay E2 specification. When all three conditions hold, the two tests reach the same coefficient and the same standard error up to small-sample correction factors. When any one fails, the mixed-model test extracts more information than the strict RM-ANOVA: the continuous biomarker contributes within-stratum variation, the AR(1) covariance is modeled directly, and the continuous exposure-decay predictor draws on the off-drug trajectory.

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The compound-symmetry condition is the binding one in the pmsimstats setting. Equation (8) presupposes sphericity: the within-subject covariance matrix is compound-symmetric. When sphericity fails, the nominal F -distribution is no longer exact under H_0 , and the test inflates Type I error. Two diagnostics and two corrections are standard. Mauchly's test³⁷ is a likelihood-ratio test for compound symmetry against an unstructured alternative; it often rejects in moderate samples but is not informative about the direction of departure. Box's ϵ , with $\epsilon = 1$ at compound symmetry and $\epsilon \geq 1/(T - 1)$ at maximum departure, is a scalar measure of departure from sphericity. The Greenhouse-Geisser correction⁵ estimates $\hat{\epsilon}$ from the observed within-subject covariance and adjusts the degrees of freedom: $F_{AB} \sim F(\hat{\epsilon} \cdot df_1, \hat{\epsilon} \cdot df_2)$ under H_0 . The Huynh-Feldt correction⁴ is a less-conservative variant recommended when $\hat{\epsilon} > 0.75$. The contemporary sphericity-violation literature^{3,6} documents substantial Type I error inflation when sphericity violations go uncorrected, with the conventional rule favoring Greenhouse-Geisser when $\hat{\epsilon} < 0.75$.

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The pmsimstats DGP imposes AR(1) within-factor correlation, which is not compound-symmetric. Under AR(1), Box's ϵ is strictly below 1, and treating the full multi-timepoint series as repeated within-subject levels under classical RM-ANOVA inflates Type I error unless a degrees-of-freedom correction is applied. The relevant magnitude is recorded in the project's own audit (docs/06-ar1-residual-correlation.tex): at $T = 8$ timepoints, fitting a mixed model with no residual-correlation structure (the naive analogue of treating AR(1) data as exchangeable) produces Type I error of 13 to 17 percent for the crossover (CO) and open-label (OL) designs, which switching to nlme::lme with corCAR1 returns to nominal. We stress that this 13-to-17-percent figure is for the unstructured/naive fit, not for the strict RM-ANOVA arm evaluated in this paper: that arm pre-averages each phase to a single value (the 2×2 form of §2.2), so its within-subject covariance is 2×2 , sphericity holds by construction, and its

Type I error is at nominal (§5). The sphericity penalty is therefore a property of the un-pre-averaged classical analysis; for the pre-averaged strict RM-ANOVA the operative cost is dichotomization, not sphericity. Either way the linear-mixed analysis with corCAR1 avoids both costs, modeling the AR(1) structure directly rather than absorbing it via a degrees-of-freedom penalty.

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For the founding pmsimstats interaction question², the strict-RM-ANOVA test is therefore a conservative reduction of the available analysis rather than a competing alternative. Where a reviewer requests ‘RM-ANOVA results’, three responses are defensible: (a) point at the existing linear-mixed analysis as a continuous-time variant of the RM-ANOVA family, (b) report the strict-RM-ANOVA test from equation (8) as a sensitivity comparator under dichotomized biomarker and binary phase, or (c) report both with explicit power-loss tabulation. The simulation study in §4–§5 operationalizes option (c).

3.3.1 Power calculation under the closed form

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For the 2×2 case, equation (8) yields a closed-form power formula that we shall use as the analytic anchor for the simulation study in §4–§5. Under the alternative $H_1 : \hat{L} = \Delta_0$, the statistic $F_{AB} = nT \hat{L}^2 / (4\sigma_e^2)$ follows a non-central F distribution with non-centrality parameter (here Δ_0 denotes the true value of the interaction contrast under the alternative, an effect size on the raw outcome scale, and should not be confused with the standardized bias δ used in the companion calibration manuscript³⁸)

$$\lambda = \frac{nT \Delta_0^2}{4\sigma_e^2}, \quad (12)$$

and power at level α is

$$1 - \beta = \Pr \left[F_{AB} > F_{1,2(n-1),1-\alpha} \mid \lambda \right], \quad (13)$$

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computable via standard non-central F functions (`stats::pf` in R with the `ncp` argument; `PROC POWER` in SAS). For sample-size calculation, equation (12) inverts to $n = 4\sigma_e^2 \lambda / (T \Delta_0^2)$, with λ chosen to deliver the target power at the design α . The formula assumes (i) sphericity, (ii) categorical biomarker at the dichotomized cut-point with equal stratum sizes, (iii) no carryover, (iv) no within-treatment relapse, and (v) i.i.d.

513 within-phase observations after within-subject differencing. All five assumptions
514 are violated to varying degrees in the pmsimstats DGP, as we shall develop in the
515 next subsection.

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518 We note that these five assumptions are not independent: violations of sphericity
519 (i) and of the i.i.d. within-phase assumption (v) are, in the AR(1) setting, two faces
520 of the same departure from compound symmetry. The analytic power formula
521 should therefore be understood as an upper bound on the power available from the
522 strict-RM-ANOVA test under the pmsimstats DGP, not a calibrated prediction.

523 3.3.2 Limitations of the strict classical formulation

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526 Unfortunately, the closed-form simplicity of equation (8) is bought at the price of
527 five strong assumptions, each of which is empirically problematic in the pmsimstats
528 setting. We present them in approximate order of severity.

- 529 1. *Categorical biomarker.* The pmsimstats biomarker (resting SBP in the prazosin-PTSD
530 application) is continuous and has no defensible cut-point. Dichotomizing at the median
531 or at a clinical threshold loses the within-stratum biomarker variation, attenuates the
532 interaction estimator, and inflates Type II error. In the carryover-sensitivity production
533 data the median split gives a $\hat{L}$ standard error roughly 1.4 to 1.7 times that of the
534 continuous- biomarker linear-mixed coefficient, translating to a 30 to 50 percent power
535 penalty.
- 536 2. *Sphericity.* Compound-symmetric within-subject covariance is empirically rejected
537 for time-series symptom data: nearby timepoints are more correlated than distant
538 ones. The AR(1) structure pmsimstats implements explicitly violates sphericity. After
539 Greenhouse-Geisser correction, the test recovers nominal Type I error but loses degrees
540 of freedom and power.
- 541 3. *No carryover model.* Strict RM-ANOVA has no provision for residual drug effect during
542 the off-drug period; the model treats off-drug observations as if no drug had been ad-
543 ministered. Under the pmsimstats DGP with $t_{1/2} = 1$ week, this misspecification biases
544 $\hat{L}$ toward zero (attenuation by 30 to 35 percent in the production data, comparable to
545 the E1 specification evaluated in the carryover-sensitivity manuscript).
- 546 4. *Balanced design required.* Equations (6) and (8) assume equal stratum sizes and equal
547 numbers of within-phase timepoints. Real trials have missing data; classical RM-
548 ANOVA handles missingness only through complete-case deletion. The MMRM frame-
549 work's likelihood-based handling of missing-at-random (MAR) data³⁶ is genuinely better-
550 behaved.

551 5. *Equal time spacing.* Strict RM-ANOVA treats time as a categorical factor with im-
552 plicitly equal-spaced levels. Real N-of-1 trials have unequally-spaced visits; pmsimstats
553 implements continuous-time AR(1) (`corCAR1`), which the strict-RM-ANOVA framework
554 cannot accommodate.

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557 These limitations motivate retaining the linear-mixed analysis with continuous-
558 time AR(1) residual correlation as primary, and reporting the strict-RM-ANOVA
559 closed-form F -statistic as a sensitivity comparator. The five violations are not
560 equally consequential: the categorical-biomarker and no-carryover assumptions
561 are the most problematic in the pmsimstats setting, together accounting for a 30
562 to 50 percent attenuation of the interaction estimator relative to the linear-mixed
563 analysis. The recommendation operationalized in §6 follows this strategy.

564 4 Cycle-by-period design grid

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567 The cycle-by-period design grid that varies the number of treatment cycles, on-
568 drug and off-drug period lengths, and on/off symmetry around the² hybrid refer-
569 ence is specified in `docs/27-cycle-period-design-sweep-plan.md` and consti-
570 tutes the second axis of the joint design-by-test sensitivity question framed in §1.
571 The empirical sweep across this grid is deferred to a follow-up paper. We shall
572 note here, in the interest of transparency, that the present paper narrows its scope
573 to the test-procedure axis at the prazosin/PTSD reference design (Hendrickson
574 hybrid open-label plus blinded discontinuation, OL+BDC; $N \in \{25, 35, 70, 100\}$,
575 $\beta_{bm} \in \{0, 0.45\}$), which the simulation infrastructure does support at production
576 rep counts. The §6 recommendation is correspondingly scoped to test-procedure
577 choice rather than the joint (cycle count, period length, test procedure) optimum
578 that the design-by-test sensitivity question requires. The follow-up paper execut-
579 ing the cycle-by-period sweep is the natural extension and is recorded as a portfolio
580 item.

581 5 Simulation design

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584 The simulation program follows the ADEMP framework of¹. The primary esti-
585 mand is the biomarker-treatment interaction inference under each of the five test
586 procedures. Four operate on the continuous biomarker and share the same inter-
587 action estimand $\beta_{bm:D}$ on the symptom scale: a within-subject treatment-effect
588 contrast regressed on the continuous biomarker (the dichotomization-free analogue

of the two-sample t -test of equation (11)); the linear-mixed-effects Wald t on $\hat{\beta}_{bm:D}$; the GEE Wald z on $\hat{\beta}_{bm:D}$ with naive sandwich variance; and the GEE Wald z with Mancl-DeRouen small-sample sandwich variance per³⁴. The fifth, the strict classical RM-ANOVA F -statistic, operates on the median-dichotomized biomarker and so targets a dichotomized estimand on a different scale. The continuous-biomarker contrast is included specifically to separate the cost of dichotomization (strict RM-ANOVA versus the contrast, both pre-averaged to one value per phase) from the cost of the test procedure proper (the contrast versus the longitudinal linear-mixed and GEE analyses). Primary inferential outputs are power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and Type I error under $\beta_{bm:D} = 0$. Secondary estimands, reported for the four coefficient-bearing arms, are bias of $\hat{\beta}_{bm:D}$ relative to a high-precision Monte-Carlo reference value of the estimand, the empirical and the mean model-based standard error, and nominal-95% CI coverage. Strict RM-ANOVA is excluded from these coefficient-based measures because its estimand is on the dichotomized scale.

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Performance measures are reported with MCSE alongside every estimate, following Morris et al.'s recommended discipline. Power and Type I error MCSE follow the binomial-proportion form $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$. Bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard MCSE of the within-replicate mean estimator.

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The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at [analysis/scripts/test-procedure-design-sensitivity/00-ade](#) which fixes the test-procedure times design-grid factorial across three studies (Study 1: test-procedure comparison at fixed design; Study 2: cycle-by-period grid at fixed test, 240 cells; Study 3: joint design-by-test heatmap, 240 cells) before the production runs commit. The present paper executes a revised Study 1; the executed design departs from the pre-registered Study 1 on several factor levels, and we document every departure in Table 1 below in the interest of full ADEMP transparency.

Table 1: Pre-registered versus executed Study 1. Departures are recorded for transparency; the executed superset of sample sizes (which restores the pre-registered N levels and adds $N = 25$) and the added continuous-biomarker contrast arm respond to referee comments on the first internal draft.

Factor	Pre-registered	Executed (this paper)
Test procedures	3 (RM-ANOVA, LME, GEE-MD)	5 (adds naive GEE and the continuous-biomarker contrast)
Biomarker effect β_{bm}	{0, 0.3, 0.6}	{0, 0.45} (the project-wide reference effect size)

Factor	Pre-registered	Executed (this paper)
Sample size N	{35, 70, 100}	{25, 35, 70, 100} (superset; adds $N = 25$)
Replicates per cell	1000 (5000 for Type I cells)	1000 power, 2000 Type I
Carryover half-life	$t_{1/2} = 3$ days	$t_{1/2} = 0$ (no carryover at the fixed reference)

Units. Carryover half-lives $t_{1/2}$ are quoted in days in this table, following the pre-registration and the main-effect manuscripts of this series. The interaction-focused manuscripts quote $t_{1/2}$ in weeks on the canonical grid {0, 0.5, 1.0}; the conversion is 1 week = 7 days. Timepoints t are weekly in both conventions.

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In the scope of the present paper (test-procedure axis only at the reference design), the simulation runs on the pmsimstats package data-generating process under the mean-moderation architecture (mean moderation), exercised through `generateData()` and the standard `lme_analysis()` fit, rather than on the reduced implementations/simple substrate. The factorial comprises $n_{\text{reps}} = 1000$ per power cell and 2000 per Type I cell, across the 40 cells defined by procedure $\in \{\text{strict RM-ANOVA, continuous-biomarker contrast, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times \beta_{bm} \in \{0, 0.45\} \times N \in \{25, 35, 70, 100\}$. Execution uses `parallel::mclapply` with `RNGkind("L'Ecuyer-CMRG")` and `mc.set.seed = TRUE`, so each replicate draws from an independent, reproducible random-number stream. The Mancl-DeRouen correction was implemented inline (no `geesmv`-style external dependency in the production path) by inflating each cluster's residual via $(I - H_{ii})^{-1} r_i$ in the sandwich meat. It produces identical point estimates and larger standard errors than the naive sandwich, and its SE for the interaction coefficient agrees with `geesmv::GEE.var.md` to within approximately 2.5% (mean inline/geesmv ratio 1.02, standard deviation 0.02, over 25 probe replicates at $N = 35$, $\beta_{bm} = 0.45$), so the correction magnitude is externally validated rather than merely plausible.

6 Results

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The simulation program was executed across the 40 cells of the five-procedure-by-effect-by- N factorial in 587 s using 7-core `parallel::mclapply` with `RNGkind("L'Ecuyer-CMRG")` per-stream seeding. Convergence was 100% across all 60{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; the Morris ADEMP cell-level summary is at `analysis/data/quick-sim/08-test-summary.txt`.

With 1000 replicates per power cell the MCSE on a power estimate near 0.5 is approximately 0.016, and with 2000 replicates per Type I cell the MCSE on a rate near 0.05 is approximately 0.005. We begin the results with the test-procedure comparison at the fixed reference design.

6.1 Test-procedure comparison at fixed design

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We note at the outset that strict RM-ANOVA targets a different estimand from the other four procedures. It operates on a median-dichotomized biomarker, pre-averaged to one value per phase, and so estimates a dichotomized-stratum interaction; the continuous-biomarker contrast, the linear-mixed model, and both GEE variants estimate the interaction coefficient $\beta_{bm:D}$ on the continuous symptom scale. An earlier version of this comparison contrasted strict RM-ANOVA directly with the mixed model and attributed the resulting power gap to the test procedure; that attribution conflated two distinct costs. We therefore introduce the continuous-biomarker contrast of §4, which differs from strict RM-ANOVA only in retaining the continuous biomarker (it shares the pre-averaging and the absence of any AR(1) machinery) and differs from the linear-mixed model only in discarding the within-phase longitudinal structure. The RM-ANOVA $\rightarrow$ contrast step therefore measures the cost of dichotomization in isolation, and the contrast $\rightarrow$ linear-mixed step measures the cost of the test procedure proper. Reading the two steps separately is the honest decomposition the earlier direct comparison lacked.

Type I error at the null cell ($\beta_{bm} = 0$), percent $\pm$ MCSE, is shown in Table 2; power at $\beta_{bm} = 0.45$ in Table 3.

Table 2: Type I error rate (percent $\pm$ MCSE) under $H_0 : \beta_{bm:D} = 0$, 2000 replicates per cell. Nominal level 5%.

Procedure	$N = 25$	$N = 35$	$N = 70$	$N = 100$
Strict RM-ANOVA (dich.)	4.55 $\pm$ 0.47	4.15 $\pm$ 0.45	5.95 $\pm$ 0.53	5.20 $\pm$ 0.50
Contrast (continuous bm)	5.25 $\pm$ 0.50	4.50 $\pm$ 0.46	6.45 $\pm$ 0.55	4.60 $\pm$ 0.47
Linear-mixed (corCAR1)	2.75 $\pm$ 0.37	2.05 $\pm$ 0.32	3.35 $\pm$ 0.40	2.90 $\pm$ 0.38
GEE (naive sandwich)	9.70 $\pm$ 0.66	8.55 $\pm$ 0.63	7.90 $\pm$ 0.60	7.40 $\pm$ 0.59
GEE (Mancl-DeRouen)	6.35 $\pm$ 0.55	5.30 $\pm$ 0.50	6.25 $\pm$ 0.54	5.90 $\pm$ 0.53

Table 3: Power (percent $\pm$ MCSE) at $\beta_{bm} = 0.45$, 1000 replicates per cell.

Procedure	$N = 25$	$N = 35$	$N = 70$	$N = 100$
Strict RM-ANOVA (dich.)	35.8 ± 1.52	46.8 ± 1.58	75.3 ± 1.36	91.3 ± 0.89
Contrast (continuous bm)	53.8 ± 1.58	67.7 ± 1.48	91.5 ± 0.88	98.8 ± 0.34
Linear-mixed (corCAR1)	56.5 ± 1.57	71.2 ± 1.43	94.5 ± 0.72	99.2 ± 0.28
GEE (naive sandwich)	67.2 ± 1.48	75.9 ± 1.35	94.5 ± 0.72	99.2 ± 0.28
GEE (Mancl-DeRouen)	57.3 ± 1.56	69.1 ± 1.46	92.7 ± 0.82	99.1 ± 0.30

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679 Four substantive findings emerge from the tables. First, and most consequentially
680 for how the comparison should be read, the power gap between strict RM-ANOVA
681 and the linear-mixed model is predominantly a cost of dichotomizing the biomarker
682 rather than a property of the test procedure. At $N = 25$ the strict RM-ANOVA
683 power is 0.358; the continuous-biomarker contrast, which differs only in retaining
684 the continuous biomarker, reaches 0.538, so dichotomization alone accounts for
685 0.180 of lost power. The further step to the full linear-mixed model raises power
686 only to 0.565, an increment of 0.027 attributable to the test procedure proper.
687 The decomposition is stable across the grid: the dichotomization component is
688 0.209 at $N = 35$ ($0.468 \rightarrow 0.677$), 0.162 at $N = 70$ ($0.753 \rightarrow 0.915$), and 0.075
689 at $N = 100$ ($0.913 \rightarrow 0.988$), while the contrast-to-linear-mixed increment never
690 exceeds 0.04. The earlier framing, which compared strict RM-ANOVA against the
691 mixed model directly and read the difference as a test-procedure effect, therefore
692 overstated the role of the test procedure by roughly a factor of five; the honest
693 statement is that a classical, sphericity-free, AR(1)-free within-subject contrast
694 on the continuous biomarker captures most of the available power, and the mixed
695 model adds a small further increment.

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698 Second, GEE with the naive sandwich is anti-conservative across the entire sample-
699 size range examined. Its null rejection rate is 0.097 at $N = 25$ and declines only
700 slowly with N to 0.086, 0.079, and 0.074 at $N = 35, 70, 100$ – still roughly $1.5\times$
701 nominal at $N = 100$. The inflation is therefore not a small-cluster curiosity that
702 disappears at trial-relevant sizes; it persists. The Mancl-DeRouen 2001 small-
703 sample sandwich correction brings the rate within or close to MCSE of nominal
704 at every N (0.064, 0.053, 0.063, 0.059), at a power cost of about ten percentage
705 points at $N = 25$ ($0.672 \rightarrow 0.573$) and seven at $N = 35$ ($0.759 \rightarrow 0.691$). The
706 corrected procedure's apparent power advantage over the linear-mixed model at

small N (Table 3) is the residue of its remaining slight over-rejection, not a genuine efficiency gain.

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Third, the secondary estimands (Table 4) locate the procedures' behavior in the standard-error calibration that the rejection rates only summarize. At $\beta_{bm} = 0.45$, the continuous-biomarker contrast and the Mancl-DeRouen GEE are well-calibrated: their nominal-95% CI coverage is 93–95% and their mean model-based SE is within a few percent of the empirical SE. Naive GEE under-covers at small N (90.0% at $N = 25$, 91.5% at $N = 35$) because its model-based SE is 9 to 12% smaller than the empirical SE, the calibration counterpart of its Type I inflation. The linear-mixed model over-covers (97.0 to 97.7%) because its model-based SE exceeds the empirical SE by 11 to 16%; this is the same conservatism visible in its sub-nominal Type I error.

Table 4: Secondary estimands at $\beta_{bm} = 0.45$ for the four coefficient-bearing arms (the interaction estimand is $\beta_{bm:D} = -2.21$, the Monte-Carlo reference value). Coverage is of the nominal-95% Wald interval; SE ratio is mean model-based SE divided by empirical SE.

Arm	Coverage $N = 25$	Coverage $N = 100$	SE ratio $N = 25$	SE ratio $N = 100$	Max rel. bias
Contrast (continuous bm)	94.9	94.8	0.99	0.98	3.0%
Linear-mixed (corCAR1)	97.7	97.0	1.16	1.11	2.6%
GEE (naive sandwich)	90.0	93.3	0.91	0.98	3.8%
GEE (Mancl-DeRouen)	94.3	94.0	1.07	1.02	3.8%

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Fourth, the comparison is not contaminated by point-estimate bias. The relative bias of $\hat{\beta}_{bm:D}$ is below 4% of the estimand $|\beta_{bm:D}| = 2.21$ for the contrast, the linear-mixed model, and both GEE variants in every cell, so the differences in power and coverage reflect variance estimation and information use rather than systematic mis-estimation. The linear-mixed procedure's Type I error sits below nominal throughout (0.021 to 0.034), the conservatism already visible in its over-coverage and its model/empirical SE ratio above one; it is consistent with corCAR1 degrees-of-freedom accounting under the short ($T = 8$) serial series. A Kenward-Roger³⁹ or Satterthwaite⁴⁰ small-sample degrees-of-freedom correction would address the conservatism and recover the small power increment it forgoes; this extension is recorded as a follow-up item.

734 **7 Design-and-test recommendation**

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737 The recommendation table below maps test procedure choice to typical chronic-
738 condition N-of-1 trial scenarios at the prazosin/PTSD reference design (Hendrick-
739 son hybrid OL+BDC, biomarker-treatment interaction estimand). The cycle
740 count and period length are held at the reference values defined in §3 because the
741 joint optimization over those axes is deferred to the follow-up paper. We present
742 the recommendation as a table of preferred (procedure, scenario) combinations
743 rather than as a single ‘best’ procedure, because the preferred choice depends on
744 features of the application that the analyst must determine. The overriding rec-
745 ommendation, however, is design-agnostic: retain the biomarker as a continuous
746 predictor, since dichotomization forfeits the large majority of the available power
747 (§5, Table 3).

Scenario	Preferred procedure	Rationale
Default (continuous biomarker, any N)	linear-mixed (corCAR1)	Highest power within nominal Type I; corCAR1 matches AR(1) physiology; well-behaved across N
Lightweight / transparent classical analysis	continuous- biomarker contrast	Within 0.03 of the mixed model on power, nominal Type I, no covariance modeling required
$N \leq 35$, concern about LME conservatism	linear-mixed (corCAR1) with Kenward- Roger	LME sub-nominal and over-covers; KR addresses this without inflating Type I
Robust-SE-required setting (e.g. unmodeled clustering)	GEE Mancl- DeRouen	Restores nominal Type I and 93–95% coverage; modest power cost
Reviewer-mandated ‘classical’ analysis	strict RM-ANOVA	Dichotomized sensitivity comparator; report alongside a continuous-biomarker analysis with the power-loss decomposition
Naive GEE (no small-sample correction)	not recommended	Type I error 1.5–1.9× nominal across all N ; under-covers

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750 The dominant message is that the biomarker should be analyzed as a continuous
751 predictor: dichotomizing it for a strict RM-ANOVA forfeits the large majority of
752 the power gap that the present study isolates, while modeling choices among the
753 continuous-biomarker procedures move power only at the margin. Within that

class, the linear-mixed analysis with continuous-time AR(1) residual correlation is the preferred default for the biomarker-treatment interaction in aggregated N-of-1 trials at the parameter scale evaluated here, with the dichotomization-free within-subject contrast a close and transparent alternative that requires no covariance modeling. Strict RM-ANOVA remains useful as a dichotomized sensitivity comparator, as the §2 closed-form derivation makes explicit, but it should be reported alongside a continuous-biomarker analysis and the power-loss decomposition rather than on its own. Naive GEE should not be used at the sample sizes examined without a small-sample correction; the Mancl-DeRouen correction restores nominal Type I error and 93–95% coverage at a moderate power cost.

8 Discussion

8.1 What the Monte Carlo study settles

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We begin the discussion by summarizing what the Monte Carlo study settles at the resolution it was run. The run on the 40-cell five-procedure factorial ($\text{procedure} \in \{\text{strict RM-ANOVA, continuous-biomarker contrast, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times \beta_{bm} \in \{0, 0.45\} \times N \in \{25, 35, 70, 100\}$, with 1000 replicates per power cell and 2000 per Type I cell) was executed in 587 s with 100% convergence across all 60{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; Morris ADEMP summary at `08-test-summary.txt`. With 1000 power replicates the MCSE on a power estimate near 0.5 is approximately 0.016 and with 2000 Type I replicates the MCSE on a rate near 0.05 is approximately 0.005. The Mancl-DeRouen 2001³⁴ correction was implemented inline by inflating each cluster’s residual via $(I - H_{ii})^{-1}r_i$ in the sandwich meat; it produces identical point estimates and larger standard errors than the naive sandwich, and its SE for the interaction coefficient agrees with `geesmv::GEE.var.md` to within approximately 2.5% (mean `inline/geesmv` ratio 1.02 over 25 probe replicates), so the correction magnitude is externally validated.

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The study resolves four substantive questions at this resolution. First, and correcting the emphasis of an earlier draft, the power gap between strict RM-ANOVA and the linear-mixed model is predominantly the cost of dichotomizing the biomarker, not a property of the test procedure. The continuous-biomarker contrast, which differs from strict RM-ANOVA only in retaining the continuous biomarker, recovers 0.180 of the 0.207 gap at $N = 25$ ($0.358 \rightarrow 0.538$ of $0.358 \rightarrow 0.565$); the further step to the full mixed model contributes only 0.027. The same decomposition holds

794 across the grid (dichotomization 0.08 to 0.21, test procedure at most 0.04). The
795 earlier direct comparison of RM-ANOVA against the mixed model therefore over-
796 stated the test-procedure effect by roughly fivefold; the corrected statement is that
797 keeping the biomarker continuous, not adopting the mixed model per se, is what
798 recovers the power.

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801 Second, naive GEE is anti-conservative across the entire sample-size range, not
802 merely at the smallest N . Its null rejection rate is 0.097 at $N = 25$ and falls only
803 to 0.086, 0.079, and 0.074 at $N = 35, 70, 100$, remaining roughly $1.5\times$ nominal
804 even at $N = 100$. The Mancl-DeRouen correction returns the rate to within or
805 near MCSE of nominal at every N (0.064, 0.053, 0.063, 0.059), at a power cost of
806 about ten percentage points at $N = 25$ ($0.672 \rightarrow 0.573$) and seven at $N = 35$
807 ($0.759 \rightarrow 0.691$). Naive GEE's nominally higher power than the linear-mixed
808 model at small N is an artifact of its residual over-rejection rather than a real
809 gain, and disappears once the size distortion is corrected.

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812 Third, the standard-error calibration of the four coefficient-bearing arms is what
813 underlies the rejection-rate differences. The continuous-biomarker contrast and
814 the Mancl-DeRouen GEE are well-calibrated, with 93 to 95% nominal-95% cover-
815 age and model-based standard errors within a few percent of the empirical stan-
816 dard error. Naive GEE under-covers at small N (90.0% at $N = 25$) because its
817 model-based SE is 9 to 12% too small, the same defect that drives its Type I
818 inflation. The linear-mixed model over-covers (97.0 to 97.7%) because its model-
819 based SE exceeds the empirical SE by 11 to 16%, the calibration counterpart of
820 its sub-nominal Type I error. All four arms are approximately unbiased (relative
821 bias below 4%), so these are pure variance- estimation effects.

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824 Fourth, the linear-mixed procedure's Type I error sits below nominal throughout
825 (0.021 to 0.034), conservative rather than nominal and consistent with `corCAR1`
826 degrees-of-freedom accounting under the short ($T = 8$) serial series. The con-
827 servatism is a variance-calibration effect, visible in the over-coverage and the
828 model/empirical SE ratio above one of Table 4, not a point-estimate effect. A
829 Kenward-Roger³⁹ or Satterthwaite⁴⁰ small-sample degrees-of-freedom correction
830 would address it and recover the small power increment it forgoes, and we record
831 this as a follow-up item; the recommendation table in §6 already lists the Kenward-
832 Roger variant for small- N use.

833 9 Conclusions

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836 In conclusion, three points are to be emphasized. First, the analyst's most conse-
837 quential choice for the biomarker-treatment interaction in aggregated N-of-1 trials
838 is to keep the biomarker continuous rather than to dichotomize it for a classical
839 RM-ANOVA. The continuous-biomarker contrast recovers about four-fifths of the
840 apparent RM-ANOVA-to-mixed-model power gap (dichotomization costs 0.08 to
841 0.21 of power across the sample sizes evaluated, while the test procedure proper
842 adds at most 0.04); the linear-mixed analysis with continuous-time AR(1) residual
843 correlation has the highest power within nominal Type I among the continuous-
844 biomarker procedures, with the dichotomization-free within-subject contrast a
845 close alternative. Naive GEE with the standard sandwich estimator inflates Type
846 I error to between $1.5\times$ and $1.9\times$ nominal across the whole $N \in \{25, 35, 70, 100\}$
847 range, disqualifying its rejection rate as a power summary; the Mancl-DeRouen
848 2001 small-sample sandwich correction restores nominal Type I control and 93–
849 95% coverage at a moderate (about 10-percentage-point) power cost, making it
850 the operational GEE comparator rather than the naive sandwich.

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853 Second, the closed-form strict-RM-ANOVA test of the biomarker-treatment inter-
854 action (§2) reduces to a two-sample t -test on the within-subject treatment differ-
855 ences under sphericity and balance. The reduction makes the strict-RM-ANOVA
856 result analytically transparent and connects the simulation findings to the classi-
857 cal split-plot literature. It also pins down what does and does not limit the test
858 here. The strict RM-ANOVA arm we evaluate pre-averages each phase to a single
859 value, so its within-subject covariance is 2×2 and sphericity holds by construc-
860 tion; consistent with this, its Type I error is at nominal (Table 2), and its power
861 deficit is wholly attributable to the median dichotomization of the biomarker, as
862 the continuous-biomarker contrast demonstrates. The sphericity penalty that mo-
863 tivates much of §2 applies instead to the alternative classical analysis that treats
864 the full $T = 8$ series as repeated within-subject levels: there the AR(1) residual
865 correlation violates sphericity and a Greenhouse-Geisser correction recovers nomi-
866 nal Type I only at a degrees-of-freedom cost, which the linear-mixed analysis with
867 `corCAR1` avoids by modeling the AR(1) structure directly. Both considerations
868 point the same way – away from a dichotomized classical analysis – but through
869 different mechanisms, and we are careful to keep them distinct.

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872 Third, the cycle-by-period design grid that constitutes the second axis
873 of the joint design-by-test sensitivity question framed in §1 is deferred
874 to a follow-up paper. The infrastructure for the sweep is in place at
875 `analysis/scripts/test-procedure-design-sensitivity/`, and the design

specification is committed at docs/27-cycle-period-design-sweep-plan.md.
The follow-up paper will deliver the joint (cycle count, period length, test
procedure) recommendation table that the present paper's §6 narrows to the
test-procedure-only axis.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

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The data and code that support the findings of this study are openly available
in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and
ADEMP-compliant cell-level summaries are versioned under `analysis/data/`;
simulation drivers are at `analysis/scripts/`. Exact paths relevant to this
manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript's simulation was executed with R 4.6.x; the project also ships a zzzcollab
Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`)
for container-based reproduction. Principal packages used by the Study 1 pipeline are
`nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `geepack`
(GEE fitting) plus an inline Mancl-DeRouen 2001 small-sample sandwich correction, `geesmv`
(independent benchmark of that correction), `data.table`, and `parallel` (`mclapply` for
the parameter-grid sweep), with `rmarkdown` plus `bookdown` for the manuscript build. The
Study 1 simulation runs on the pmsimstats package data-generating process under the mean-
moderation architecture via `generateData()` and `lme_analysis()`, not on the reduced
implementations/simple substrate.

Random-number generator. The driver sets `RNGkind("L'Ecuyer-CMRG")` and
`set.seed(20260507)`, then runs the cell-by-replicate jobs through `parallel::mclapply`
with `mc.set.seed = TRUE`, so each replicate draws from an independent, reproducible
random-number stream.

Data and code availability. Per-replicate simulation outputs are checkpointed at
`analysis/data/quick-sim/08-test-replicates.rds`; the extended Morris ADEMP
cell-level summary (rejection rate, MCSE, convergence, bias, empirical and model

SE, coverage) at the companion `08-test-summary.txt`, with the Monte-Carlo reference estimand at `08-test-betatruer.txt` and the Mancl-DeRouen benchmark at `08-test-mancl-benchmark.txt`. The driver is `analysis/scripts/test-procedure-design-sensitivity/01` (superseding the earlier `analysis/scripts/quick-sim/08-test-procedure-prototype.R`). The pre-registered ADEMP plan, and the deviations recorded in Table 1, are at `analysis/scripts/test-procedure-design-sensitivity/00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/08-test-procedure-design-sensitivity/`.

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